Received: 10 August 2020 | Revised: 11 September 2020 | Accepted: 5 October 2020

DOI: 10.1111/1556-4029.14615

Check for updates

JOURNAL OF FORENSIC SCIENCES

CRITICAL REVIEW

Criminalistics

Forensic investigation approaches of searching relatives in DNA databases

Jianye Ge PhD[1,2] | Bruce Budowle PhD[1,2]

[1]Center for Human Identification, University of North Texas Health Science Center, Fort Worth, TX, USA

[2]Department of Microbiology, Immunology, and Genetics, University of North Texas Health Science Center, Fort Worth, TX, USA

Correspondence
Jianye Ge PhD, Center for Human Identification, University of North Texas Health Science Center, Fort Worth, TX, USA.
Email: jianye.ge@unthsc.edu

Funding information
This study was supported in part by Award 2019-DU-BX-0046 (Dense DNA Data for Enhanced Missing Persons Identification), awarded by the National Institute of Justice, Office of Justice Programs, U.S. Department of Justice.

Abstract

There are several indirect database searching approaches to identify the potential source of a forensic biological sample. These DNA-based approaches are familial searching, Y-STR database searching, and investigative genetic genealogy (IGG). The first two strategies use forensic DNA databases managed by the government, and the latter uses databases managed by private citizens or companies. Each of these search strategies relies on DNA testing to identify relatives of the donor of the crime scene sample, provided such profiles reside in the DNA database(s). All three approaches have been successfully used to identify the donor of biological evidence, which assisted in solving criminal cases or identifying unknown human remains. This paper describes and compares these approaches in terms of genotyping technologies, searching methods, database structures, searching efficiency, data quality, data security, and costs, and raises some potential privacy and legal considerations for further discussion by stakeholders and scientists. Y-STR database searching and IGG are advantageous since they are able to assist in more cases than familial searching readily identifying distant relatives. In contrast, familial searching can be performed more readily with existing laboratory systems. Every country or state may have its own unique economic, technical, cultural, and legal considerations and should decide the best approach(es) to fit those circumstances. Regardless of the approach, the ultimate goal should be the same: generate investigative leads and solve active and cold criminal cases to public safety, under stringent policies and security practices designed to protect the privacy of its citizenry.

KEYWORDS
DNA database, CODIS, familial searching, Y-STR database, investigative genetic genealogy, STRs, SNPs, genotyping technologies, privacy concerns, legislation

1 | INTRODUCTION

Forensic biological samples collected from a crime scene are typically typed for autosomal short tandem repeat (STR) loci. The current core loci in the United States are composed of 20 autosomal STR (A-STR) loci, defined by the Federal Bureau of Investigation (FBI) Combined DNA Index System (CODIS), which can provide a high power of discrimination. The generated A-STR profile, whether a single source, a deconvoluted single source from a mixture, or a mixture, may be uploaded to a Local DNA Index Systems (LDIS), State DNA Index Systems (SDIS), and/or the National DNA Index System (NDIS) to search for a "direct" match (at either moderate or high stringency) referred to as a "hit." Although the 18 million offender and arrestee

The opinions, findings, and conclusions or recommendations expressed in this publication are those of the authors and do not necessarily reflect those of the U.S. Department of Justice or the above acknowledged persons.

 © 2020 American Academy of Forensic Sciences

profiles currently in NDIS [1] would appear to be quite large (~5.5% of the US population based on 327 million inhabitants as of 2018), the hit rates are only approximately 30–50% [1]. Thus, there still are a large number of unsolved cases. Indeed, there are about one million forensic profiles in NDIS (as of June 2020) [1].

To increase the number of investigative leads that can be generated through DNA testing, there are several indirect approaches to identify a potential source(s) of a forensic biological sample primarily by searching profiles of relatives of the donor of the crime scene sample, if such relatives' profiles reside in the government or private DNA database system. These methods include familial searching, Y-STR database searching, and investigative genetic genealogy (IGG). Mitochondrial DNA testing also can be used, but is not discussed herein. Each of these approaches searches either: 1) a forensic database managed by the government (e.g., NDIS [1]) or a genetic genealogy database managed by private citizens or private companies (e.g., GEDMatch [2]). All three approaches require further investigations utilizing genetic and/or nongenetic information, depending on the data and technology available. Figure 1 summarizes the general workflow for each of these three approaches. For illustrative purposes, it is assumed that a single-source profile is generated from a crime scene sample

Highlights

- Three DNA searching based investigation approaches (familial searching, Y-STR database searching, and investigative genetic genealogy) are summarized.
- These approaches are compared in terms of genotyping technologies, searching methods, database structures, searching efficiency, data quality, data security, and costs.
- The methods vary in their capability to identify relatives of the true donor of a crime scene sample.
- The privacy and legislation considerations of these approaches are raised.

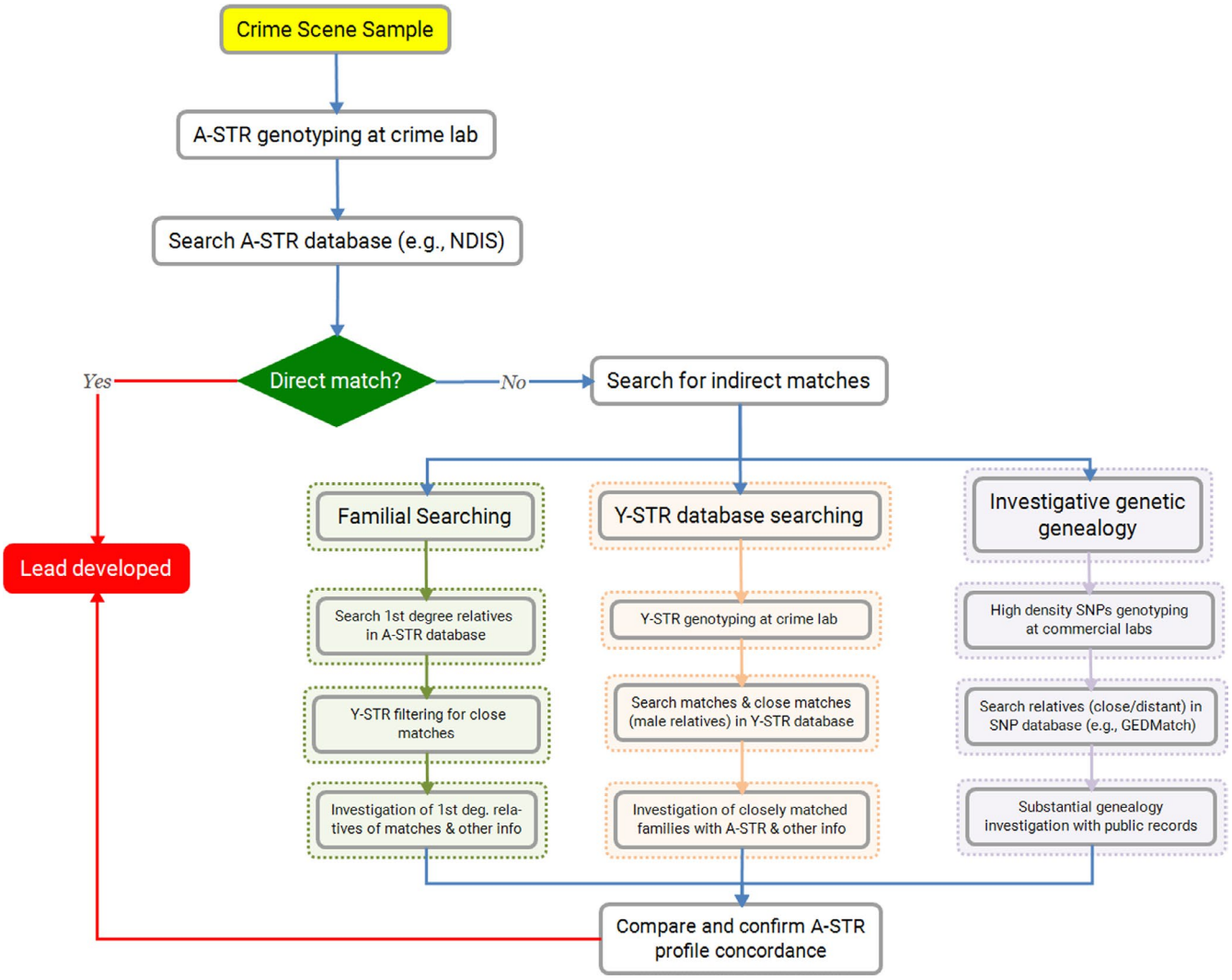


FIGURE 1 The workflow of direct and indirect DNA database searching. A-STR stands for autosomal STR and Y-STR stands for Y chromosome STR. First-degree relatives include parents, children, and full siblings

and searched in a database, although a mixture profile or a deconvoluted single-source profile from a mixture sample may also be used in a search. All three approaches have been successfully used to solve cold or active cases [3–7]. This paper describes and compares these approaches in terms of genotyping technologies, searching methods, database structures, searching efficiency, data quality, data security, costs, and raises some privacy and legal issues for further discussion.

2 | FAMILIAL SEARCHING

Familial searching was the first indirect database searching approach used to develop investigative leads, which was based on using existing government DNA database infrastructure and forensic laboratory system [8,9]. The rationale of familial searching is that a high proportion of convicted felons have close relatives who also were convicted of a crime [10]. Thus, the lead value of DNA databases, particularly local ones (i.e., LDIS and SDIS), can be extended beyond the success rates of direct matching.

In familial searching, a forensic A-STR profile generated from a crime scene sample is searched first against an offender and/or arrestee database to generate direct matches (i.e., hits). If unsuccessful, similar but not identical profiles can be ranked regarding potential first-degree kinship relationships using measures such as the likelihood ratio and/or the number of shared alleles [11–13]. Among a candidate list of the similar, but not identical, profile matches could be a close relative (parent, child, or full sibling) of the donor of the forensic profile. The top male candidates of the list (e.g., the top 100) are then subjected to Y chromosome STR (Y-STR) typing, and only those candidates with a matching Y-STR profile are considered for further investigation based on other nongenetic information (e.g., age, location, etc.). Those candidates that do not share a similar Y-STR profile with that of the crime scene sample are eliminated as they cannot be first-degree paternal relatives. Thus, potential privacy intrusion is minimized as the two-step process of A-STR typing followed by Y-STR typing minimizes the risk of including unrelated individuals. To the best of our knowledge, there has yet to be reported a false-positive male relative by familial searching. Furthermore, states in the United States that have employed familial searching generally have stringent requirements, such as an approval process, specified and limited offense types, and an affirmative assertion by specific stakeholders that all leads have been exhausted.

While familial searching can extend the investigative lead value of current DNA databases, the search is limited to first-degree relatives and typically males (note that California Department of Justice has identified female donors through a familial searching process [14,15]). One well-known example of familial searching is the Grim Sleeper case [3], in which the evidence donor (and subsequently through additional investigation the perpetrator), Lonnie David Franklin, was identified initially through an indirect match (or association) between the evidence profile and the DNA profile of Franklin's son, whose DNA profile was entered in the California SDIS because of a felony weapons charge.

The United Kingdom and New Zealand were the first countries to implement familial searching, and between 2009 and 2017, 210 familial searches have been conducted in the United Kingdom National DNA Database (NDNAD) [14,16]. In the United States, NDIS does not support familial searching, but permits each state to determine its own policy. As of 2019, at least 12 states in the United States have implemented familial searching, while only Maryland and Washington DC have explicitly prohibited familial searching [14,16,17]. According to Piquado et al. [14], two of those biggest states, Texas and California, have together only conducted less than 100 familial searches over the past decade. The limited use is likely due to stringent criteria that must be met to employ familial searching [18]. There may only be a few hundred familial searches that have been conducted globally.

3 | Y-STR DATABASE SEARCHING

Familial searching relies on the large sets of A-STR profiles in government databases and subsequent manual Y-STR typing of select candidates. Alternatively, if a Y-STR forensic database was available, a Y-STR profile generated from a crime scene sample could be searched if the direct A-STR search did not yield any hits. Depending on the number of Y-STR loci comprising the haplotype, matched Y-STR profiles have a high probability of belonging to males of the same lineage. Moreover, closely matched Y-STR profiles may have a high probability of belonging to the same male lineage of more distant relatives. Therefore, relatives of these matched Y-STR profiles can be further investigated with their A-STR profiles, if available, and other nongenetic information, as is done with familial searching. A major difference between using a Y-STR database search and the familial searching approach is that the former approach can extend the search beyond first-degree relatives.

The Netherland Vaatstra Case is one example of a successful investigation using Y-STR database searching [7]. In that case, after familial searching failed, DNA samples were obtained from 6,600 local male residents. In the first batch of Y-STR typing (n = 81 samples), there were two Y-STR profiles that matched the evidence Y-STR profile, but both were excluded by A-STR typing. While these two matching individuals had different last names, they shared the same paternal ancestor according to genealogical records. Subsequently, only those individuals who voluntarily provided DNA samples and shared a surname with either of the two matching individuals were analyzed for Y-STR. Using this information, one of the subsequent samples in the candidate list had the same Y-STR and A-STR profiles as those of the crime scene evidence. The Baiyin Case or China “Jack the Ripper” case [4,5] is another successful example, in which the DNA profile of the evidence donor's uncle was collected and entered into the forensic DNA database as a result of his bribery conviction. The uncle's Y-STR profile matched the evidence profile in the Baiyin Case

through a Y-STR database search. This information indicated a potential source of the crime scene sample, which was eventually supported with a direct A-STR profile match.

While China was the first country to implement a large-scale forensic Y-STR database [19–22], Singapore, Australia, New Zealand, and The Netherlands also have launched forensic Y-STR databases. These forensic Y-STR databases are different from the Y-STR Reference Haplotype Database (YHRD) [23]. YHRD is a population database for estimating Y-STR haplotype frequencies and research purposes, so the profiles in YHRD are anonymized. A forensic Y-STR database serves a different purpose, that is, developing investigative leads. In addition, a number of successful Y-STR searching cases have been assisted with the Y-STR data contained within public genealogy databases (e.g., https://www.geni.com/, https://www.mitoydna.org/, etc.) [24–26] and Erlich et al. [27] also described the power of Y-STRs for identifying the source of a sample.

4 | INVESTIGATIVE GENETIC GENEALOGY (IGG)

Recently, some law enforcement agencies have turned to IGG (also termed forensic genetic genealogy) to determine the identity of human remains or biological evidence primarily from cold cases. Although the 20 core A-STR loci required by CODIS are quite powerful for individualizing a sample, they do not have sufficient power to associate relatives typically beyond first degrees, and at times are even uninformative for first-degree relatives [28]. Far more genetic markers are needed to identify more distant relatives, as well as to associate siblings who by chance do not share the on average ~50% of their alleles (or greater). IGG exploits high-density SNP profiles, generated either by microarray (e.g., ~850 000 or more SNPs) or whole-genome sequencing (WGS) (e.g., millions of SNPs) technologies. A SNP profile is searched against a privately or publicly available high-density SNP profile database(s) for potential relatives in the database using measures such as the total length of the shared DNA segments between a pair of profiles [2,6,29–32]. IGG methods are able to determine 3rd degree or greater relationships with reasonable accuracies well beyond current A-STR based methods.

One of the most noted IGG cases is the Golden State Killer case [6]. In this case, a high-density SNP profile generated from a crime scene sample was uploaded to the public database GEDMatch [2]. Several of the many potential relatives' profiles in the GEDMatch search suggested that there were distant relatives (equivalent to third cousins) in the database. After potential relatives were associated via genetic data, the genealogy part of the investigation began. With substantial searching of genealogical records anchored on these potential distant relatives, together with other information (such as age and location), investigators were able to narrow their search to a particular person. Then, the A-STR profile(s) derived from the evidence and from the suspect were compared and deemed a match.

The United States was the first country to apply IGG, likely because of the Direct-to-Consumer (DTC) market and the existence of GEDMatch, a database populated voluntarily with high-density SNP profiles from a portion of the subscribers of the DTC genetic testing services. GEDMatch was recently acquired by Verogen [33]. As of May 2020, there have been at least 109 positive associations to both missing persons and suspects through GEDMatch by Parabon NanoLabs alone [34]. European countries [35–37] and China [38,39] also have started to consider IGG for forensic investigations.

5 | GENOTYPING TECHNOLOGIES

Both familial searching and Y-STR database searching rely on STR profiles and either initially or subsequently A-STR typing, respectively. An A-STR reference profile usually contains 20 CODIS core loci with a few additional common STR loci, and a Y-STR profile may contain 23–40 Y-STR loci, depending on the commercial kit. The amplified STR fragments are usually analyzed by Capillary Electrophoresis (CE) (e.g., 3500xl Genetic Analyzer). Using accompanying software, such as GeneMapper IDX, STR alleles are called based on the length of the DNA fragments and specific dye labels. Recently, the massive parallel sequencing (MPS) technologies, such as Illumina's MiSeq and Thermo Fisher's S5, also can analyze and call STR alleles with higher resolution (i.e., underlying sequence data) and greater throughput. MPS of STRs is backward compatible with CE-based STR typing results contained in DNA databases worldwide, as sequence data can be converted to length-based allele data. The recommended optimum DNA input amount for STR typing technologies is approximately 500–1,000 pg, but a full DNA profile may be obtained with less DNA input amounts [40–44].

IGG requires high-density SNP profiles, which are generated by microarray or WGS technologies. The major DTC genetic testing companies (e.g., 23andMe, Ancestry.com, MyHeritage) generate their own SNP data, while the major IGG services (such as Parabon NanoLabs, Bode Technology, DNA Doe Project) usually outsource SNP genotyping to private service laboratories. These service laboratories usually use microarrays to generate SNPs [31], as the microarray genotyping is relatively inexpensive, accurate (~99.9% genotyping accuracy [45]), and can provide sufficient information to determine distant relationships with varying degrees of confidence. WGS can provide a much higher density of SNPs (e.g., >10 million) compared with microarray technologies. However, currently, only a subset of these SNPs can be used in an IGG search, because most of the profiles in the GEDMatch and other major genealogy databases were genotyped via microarrays at less than 1 million SNPs. Microarrays require much more input DNA than CE STR genotyping. Bode Technology recommends using at least 20 ng of DNA [46]. Greytak et al. [31] summarized the cases processed by Parabon NanoLabs and found that 77.3% of samples had >2.5 ng DNA. While WGS may use less than that of microarrays, the amounts necessary for complete profiles with high accuracies are not published at this time.

Although both microarray and WGS technologies are not validated by the forensic laboratories, nor is there published supportive

data released by the current service providers for those processes, they are being used as investigative tools. Currently, the US Department of Justice requires direct A-STR matching confirmation of any association made by IGG [47-49]. This confirmation approach, when applicable, might obviate some of the validation requirements for IGG, although privacy, sample consumption, and potential probable cause issues may need to be considered. Additionally, while important, this is an unwieldy recommendation in missing persons cases since the vast majority of these cases rely on indirect associations to support an association or an identification. Therefore, it is crucial that validation studies should be performed on these methods so that the reported associations to stakeholders and customers are appropriately and scientifically supported. In addition, validation studies in general define the limitations of the method and thus better inform the customer of the potential risks associated with this method, such as consumption of sample and uncertainty associated with predicted kinship relationships.

6 | SEARCHING METHODS

The core method of the familial searching approach is to determine whether two A-STR profiles are from parent–child or full-sibling relationships, or are from unrelated individuals (unrelated herein is any relationship beyond first-degree relatives). There are two genetic-based methods that could be used for assessing potential relationships, Identity-by-State (IBS) or likelihood ratio (LR) [12,13,50]. The IBS method simply counts the number of shared alleles across all loci between a pair of profiles. If the number of shared alleles reaches or exceeds a threshold value, the donor of the sample in the database is considered a candidate. In general, a high IBS indicates a potential close relationship. The IBS method does not depend on allele frequencies, which makes it simple to implement and robust with complex populations. However, IBS does not make full use of the genetic data. In contrast, the LR method compares the likelihood of the profile pair given two hypotheses: 1) They are related (parent–child or full sibling), and 2) they are unrelated. The LR method is more often used in familial searching, because of its use of population allele frequencies, and it provides a higher accuracy compared with the IBS method. If there are multiple possible populations, multiple LRs may be calculated for all reasonable populations, and the range of LRs or the minimum LR is provided [11].

The Y-STR database searching approach is used to determine whether two Y-STR profiles are from the same male lineage, either as an exact match with all alleles shared at all typed loci, or as a partial match with one or a few mismatches. More distant male relatives have a higher chance of mismatches and yet still be paternally related. The total number of mismatched alleles or repeats different can be counted, and if it is less than some predefined thresholds, the profiles are considered as related candidates to be followed up with further investigation [20,51]. The mismatch threshold for placing candidates in the related and unrelated categories depends on the set of Y-STR loci and the mutation rates of the loci, which will vary across different populations [52-54]. The LR method may also apply to Y-STR haplotype comparisons given the same hypotheses as described above [55]. However, there could be many different relationships within one male lineage (ranging from father–son to distant cousins), which can lead to different LRs.

An IGG database search usually determines the relationships between a pair of high-density SNP profiles by measuring, for example, the total length of shared IBD segments in centiMorgans (cM) [2,32,56,57], and many potential relatives can be identified by an IGG search. In general, the higher the total length of shared IBD segments is, the closer the relationship is. Methods based on the numbers of shared alleles [58] or rare alleles [59] also may provide high accuracies, but are not implemented by the major IGG databases. These databases usually only use the autosomal SNPs in relationship calculations, although the SNPs on X, Y chromosomes, and mitochondrial genome may be stored in the databases (personal communication, Dr. Colleen Fitzpatrick), possibly because distant relatives may not share the same lineage markers.

7 | DATABASES AND SEARCHING EFFECTIVENESS

7.1 | Familial searching based on A-STR DNA databases

Globally, more than 60 countries have implemented governmental forensic DNA databases, and worldwide there are over 125 million offender/arrestee profiles in these databases [60]. The NDIS in the United States contains over 14 million offender profiles and about 4 million arrestee profiles as of May 2020 [1], the United Kingdom National DNA Database (NDNAD) has about 6.5 million profiles as of March 2020 [61], and China holds the largest DNA database with about 68 million profiles as of 2017 [60].

Although the NDIS dataset seems to be of large size, it only covers up to ~5.5% of the US population. Assuming an average family size of 4 (i.e., parents and 2 children), all profiles in the forensic DNA database are from different families, the accuracies to determine first-degree relatives are 100%, and all populations are represented according to census data, familial searching could cover up to 22% of the population, although none of these assumptions actually hold. Some profiles in the databases are from the same nuclear family, and the accuracies of determining a pairwise first-degree relationship are not 100% effective due to the limited number of preselected STR loci used in these databases and the stochastic distribution of alleles from parent to offspring.

The US CODIS requires testing with 20 core A-STR loci [62], the China National DNA database recommends 23 core A-STR loci [63], and the European Union recommends 16 European Standard Set (ESS) A-STR loci [64]. However, profiles uploaded to the database years ago may contain fewer loci due to older policies and capabilities, and crime scene evidence may be compromised, resulting in DNA profiles containing less than the prescribed core A-STR

loci. Therefore, a good portion of pairwise comparisons by familial searching may only measure the kinship relationships based on a subset of these loci (e.g., in the United States the original 13 CODIS core loci [65] or less for crime scene profiles).

With these limited number of STR loci, the accuracy of determining a full-sibling relationship is hampered and much lower than that of a parent–child relationship. Consider the original 13 CODIS loci and a LR threshold of 1,000, the false-negative rates of determining true parent–child and full-sibling relationships are 14.1% and 42.5%, respectively, and the false-positive rates of determining unrelated as parent–child or full-sibling relationships are 0.0132% or 0.093%, respectively [13]. For a database with 1 million profiles, ~1,000 closely matched but unrelated profiles (parent–offspring and full-sibling combined) may be included on the candidate list. The false-positive rate, although low, has an additional filter of Y-STR typing that reduces the risk to nominal levels. However, the false-negative rate means that truly related individuals that are in the database will not be identified at a reasonably high rate.

Moreover, investigating hundreds of families may not be practically feasible. Therefore, laboratories usually only pick top candidates (ranked by LR) to fill one or two 96-well plates for subsequent Y-STR analysis [36], as the current US practices do not typically type Y-STR profiles for convicted felon and arrestee reference samples. Because the majority of these candidates are males [22], filtering with highly discriminating Y-STR profiles (e.g., a discrimination power of ~1 in 1000(s) for the major populations [66]) is able to efficiently exclude most, if not all, of these unrelated candidates. With both A-STR searching and subsequent Y-STR filtering, only close male relatives (i.e., father, son, and brother) can be potentially identified through familial searching. Female relatives also may be identified in a similar manner, if mitochondrial DNA is used in filtering [14,15].

7.2 | Y-STR database searching

A number of countries are building national or regional Y-STR databases. China was the first country to develop a Y-STR database in the early 2010 s. Similar to the A-STR databases, the China Y-STR database contains both reference (e.g., offender, arrestee) and forensic profiles. The Netherlands created its Dutch Y-STR database for criminal investigations around 2012 [7]. Singapore launched its national Y-STR database in June of 2018 by retyping all the convicted offender reference samples (personal communication, Dr. Christopher Syn). For casework in Singapore, Y-STR analysis is conducted if there is no A-STR match in the database or if a male component is detected in a mixture sample (e.g., samples from sexual assault cases). New Zealand has been operating a national Y-STR database since October 2018 (personal communication, Dr. SallyAnn Harbison). Due to financial limitations, no reference samples were genotyped by Y-STRs and only crime scene Y-STR profiles are included in that database. Australia also has a national Y-STR database (personal communication, Ms. Carole Field), which includes thousands of Y-STR profiles generated from offenders, suspects, crime scene samples, and unidentified deceased persons. The databases in these countries do not associate the DNA profile with genealogy information in advance, but such additional information is necessary if a match is obtained. In the United States, Missouri State Highway Patrol Crime Laboratory also has developed a local Y-STR database by cumulating the profiles through casework (personal communication, Mr. Brian Hoey).

To appreciate the efficiency of the Y-STR database searching approach, a scenario is presented in which all male lineages are not related to each other and one male in each lineage is selected as the representative of each lineage. If a male lineage on average contains 20 males, then only 5% of males in a population are needed to cover all males in the entire population (namely, one male can serve as a representative for all 20 males in the same lineage), although a higher percentage would need to be collected to overcome sampling error. In reality, depending on how a male lineage is defined, the average number of males in a lineage can be higher or lower, and likely many male lineages may be related to some degree. Thus, it is highly likely that the entire male population could be covered by a smaller proportion of representative males. In this regard, Y-STR database searching can be considered to be more effective than familial searching with A-STR databases (and subsequent Y-STR filtering) to identify potential relatives because more distant lineage relationships may be detected [20,22,67]. From a regional or local perspective, Y-STR database searching could be quite efficient in rural areas as these areas tend to have less migration and thus are relatively stable regarding population structure than that of large cosmopolitan areas.

Theoretically, the Y-STR database searching can cover all male relatives of the profiles in the database, but it would depend on the degree of relationships and mutation rates of the markers. The power of discrimination (PD) of a Yfiler Plus profile is close to 1 in 10,000 [20], depending on the population, which means that almost 99.99% of unrelated males would be excluded in a Y-STR database search. In general, adding more Y-STRs should increase the discrimination power of a profile to better exclude unrelated males. However, due to linkage disequilibrium among Y-STRs, the discriminating power increases gradually at best as more markers are added to the haplotype. Moreover, mutations of the Y-STRs within the same lineage over generations will generate a certain degree of mismatches that will need to be considered to minimize false exclusions. Liu et al. [20] estimated that a pair of Yfiler Plus profiles with less or equal to 5 mismatched repeat steps are more likely related, if only male relatives up to 4 generational steps are considered. This threshold should vary with the Y-STRs in a profile (both the number of loci and mutation rate of each locus), the population, and specific relationships. As the number of Y-STRs included in a kit increases, the cumulative mutation rate across the Y-STR haplotype also increases. Therefore, it would be more challenging to differentiate related and unrelated categories for a profile pair with several mismatches (e.g., 4 ~ 6 total mismatched steps). In addition, the accuracy of identifying

male relatives by Y-STR database searching decreases with the distance of the relationships, which is the same as for any relationship estimation method.

Based on the experiences in China and the study by Liu et al. [20], relationships within four meioses/generations could be identified with high confidence by Y-STR database searching. Two unrelated profiles have >5 mismatched steps in most cases (i.e., >99.975%), and almost all close relatives are within 4 meioses have less than ≤5 mismatched steps (i.e., 99.999878% for first cousin and higher accuracies for closer relatives) [51]. More distant male relatives also can be identified to develop more distant investigative leads compared with familial searching, although substantial genealogy investigation may be needed. One notable limitation of Y-STR database searching is that it does not inform with maternal relationships. However, mitochondrial DNA sequencing could provide maternal lineage information, if so desired. Since the majority of the crimes (and particularly violent ones) are committed by males [68,69], the Y-STR database should be able to assist in the majority of criminal cases of interest, particularly for a number of sexual assault cases in which A-STR typing is uninformative and a single-source Y-STR profile can be generated.

7.3 | Genetic genealogy databases and IGG

As of 2019, there have been more than 26 million SNP profiles generated by the DTC companies, and the number is increasing [70]. In addition, there are at least 22 individual genealogy databases in the United States and Europe alone [71]. One of the most widely used databases for forensic investigations is GEDMatch, which allows users to voluntarily upload their DNA profiles primarily to find relatives. Around 1.4 million profiles have been uploaded to GEDMatch as of 2020. However, GEDMatch updated its policy in 2019 to require users to decide to opt in for law enforcement to search their profiles, and only ~260 k users opted in as of June 2020 [2]. FamilyTreeDNA also is actively working with law enforcement investigating cold cases by providing access to its 1.3 million profiles [72,73]. FamilyTreeDNA has invoked a default opt-in policy for access to the profiles that can be searched by law enforcement [74]. Other DTC companies are not partnering with law enforcement and may only do so if served a search warrant. Considering only GEDMatch and Family Tree, hundreds of cases have been assisted by IGG [30,31,34]. For example, Parabon NanoLabs alone has been involved in ~500 cases and generated at least 109 positive identifications of suspects [34]. However, the number of incorrect or false leads is unknown.

With the current large number of SNPs and numbers of profiles in the genetic genealogy databases, IGG is able to identify rather distant relationships (e.g., 6th degree or more [31,32,56]) for both male and female relatives. According to Erlich et al. [27], 60% of the US populations can be identified through a genetic genealogy database with 1.28 million profiles. Theoretically, a database containing 2% of a population can provide a third-cousin match to any person in the target population. However, in reality, the majority of the profiles in the current databases are of European descent [27,75], and thus, Caucasian individuals are more likely matched through a current high-density SNP search.

Although many matches may be obtained through a database search, more effort is required during the follow-up investigation of public records to determine the source of the crime scene sample(s) [76]. Many of matches could be very distant relatives and the number of distant relatives of an individual identified increases exponentially (e.g., almost 1000 fourth cousins and close to 5000 fifth cousins [77]). For each match, a few to thousands of hours of public record research may be needed to confirm the association [76], depending on the distance of the relative and the amount and quality of the information in the public records. Most genealogy investigations focus on third degree or closer relationships because of the substantial burden of searching fourth degree or more matches. An efficient way to improve public record searching is triangulation, which identifies a source by intersecting between two potential families [31,78]. Other information, such as location and age, also can be very helpful to increase success during genealogical surveying.

Y-STR, Y-SNP, X-SNP, and mitochondrial DNA SNPs may be included in the genealogy databases, but these data are usually not used in typical IGG searching. In certain scenarios, Y-SNP or Y-STR haplotypes may be used in association with population affinities (e.g., https://yfull.com/tree/) and/or surnames to facilitate the investigation, and a number of successes have been reported [24–26,79].

8 | COSTS

The A-STR and Y-STR genotyping have the benefits of being performed utilizing existing equipment in forensic laboratories. The major cost of equipment is the CE instrument, around ~$200 k in the United States. The cost of A-STR or Y-STR genotyping per reference sample in the United States is around $30-50 per profile excluding labor cost, depending on workflow and the reagents used (costs estimated from the University of North Texas Health Science Center, Center for Human Identification). The costs of analyzing a reference sample on average are less than that of casework samples, especially if some of the steps in the workflow (e.g., extraction, quantification) are skipped. In the scenario that a sample is typed for both A-STR and Y-STR, extracted DNA can be used to type both sets of markers (or direct amplification can be performed). Thus, the total cost should be lower than two independent tests (including DNA extraction). The cost also could be lower if a kit enables typing both A-STR and Y-STR simultaneously.

For a familial searching case, if one or two 96-well plates of samples were typed for Y-STR filtering, the subsequent cost would be a few thousand dollars. In contrast, if a database contained Y-STRs, the familial investigation based on a Y-STR match requires A-STR genotyping of several male relatives of the match, which may cost a few hundred dollars, presuming these A-STR profiles are not already in the database. The upfront cost of building an independent

Y-STR database may be high (comparable to building an A-STR database), if all offender/arrestee samples are retyped (such as was done in China and Singapore for their Y-STR databases). An alternate approach would be to add Y-STR profiles going forward and supplement the database with Y-STR reference profiles through regular casework, such as sexual assault cases; there is little cost associated with uploading an already generated a profile. In addition, a familial investigation also may be needed for matches from the Y-STR database searches, which could add additional costs to Y-STR database searching. Adding Y-STRs to government databases would facilitate both familial searching and Y-STR database searching.

Generating high-density SNP profiles requires additional expensive equipment (e.g., Illumina iScan, NovaSeq, etc.), at the cost of $100–$1000 k. In addition, servicing and maintaining WGS-MPS instrumentation can be costly. It may be unrealistic for a regular forensic laboratory to purchase and maintain such equipment. Therefore, SNP genotyping is and will likely continue to be outsourced to the genotyping service laboratories. The cost per sample is about $100 to a few $100 with microarray typing and $1000 or more for WGS. If an IGG case is outsourced to a genealogy service provider, the SNP genotyping is included in the package and the total cost can be $3000–$10 000 or more, as the majority of the cost is associated with the genealogy investigations of public records.

9 | DATA QUALITY

The genotype profiles in national DNA databases were generated with validated technologies in accredited forensic public and private laboratories who also are subject to proficiency testing with the specific methodologies (note that private laboratories are not permitted to upload data to CODIS). While different kits may be used by different laboratories, the discordance rate between the genotypes generated is low and typically at most at a single locus in a profile [80,81]. Additionally, STR genotypes generated from CE and MPS have low discordance rates [44]. The high-density SNP profiles in the nongovernment databases generated by microarray or WGS technologies have not been forensically validated (i.e., no peer-review publications describing the studies), and the service laboratories may not have participated in proficiency testing on the particular methodology. The discordance rate of the genotypes generated from different platforms could be higher than those between CE kits (e.g., 0.34% [82]).

Although genotyping errors may occur, the sheer numbers of SNPs may make the error rate moot, except possibly for the most challenged of samples. A-STR genotyping using forensically validated procedures is used to confirm or refute an association derived by IGG. Therefore, while genotyping errors, or more likely genealogy errors, may lead to false exclusions or inclusions, false identifications should not be made if followed up with proper A-STR typing of the alleged source and the crime scene evidence. However, the degree of error and quality of data that are needed to minimize practical error in inferring a kinship relationship have not been defined, and validation studies are essential and necessary to address this threshold for SNP profile comparisons.

Identifying information of the offender/arrestee profiles in the forensic A-STR and Y-STR databases is maintained separately from the DNA data. While the contact information (i.e., name and email address) are available, for example, in GEDMatch, they also may contain inaccurate, vague, or false information (e.g., an alias instead of an actual name, a common name, an anonymous email, a name from a representative instead of an actual user), which could hamper an investigation [31].

10 | DATA SECURITY

The national DNA databases are managed by government agencies under stringent security and privacy policies and laws [83]. Only the approved forensic laboratories can upload, download, and search these databases. In contrast, GEDMatch, for example, is publicly accessible and has less security stringency than national DNA database systems. Two recent studies [84,85] have demonstrated how DNA profiles in GEDMatch may be revealed and how fake profiles may be uploaded to the database to impede forensic investigations. Other DTC genetic databases may be more secure, as they are not open for searching by the public. While all databases may be vulnerable to hacking, unlike some IGG databases [86,87], CODIS has not had any security breach to date. In addition, the data in the forensic A-STR and Y-STR databases cannot be shared with external entities, except as described by the DNA Identification Act or by court order. However, the SNP profiles within the DTC companies may be shared with third parties, such as research communities or pharmaceutical companies, either for research purposes or business reasons. Additionally, DTC companies, together with the data, can be sold to investors [88], and the data security policy may vary because of ownership change.

11 | PRIVACY CONCERNS

Privacy concerns, especially those associated with genetic data and database searching, are issues often discussed and debated. This paper does not attempt to examine these issues in any depth. Instead, it seeks to merely raise some of the potential issues so that scientists, stakeholders, and the public can consider them in evaluating the use of the above-described approaches. All three approaches can extend investigations by searching for potential relatives of the sample donor. Thus, they each have greater outreach than just individuals contained in the database. As the list of identified relatives increases, more individuals unrelated to the crime may become part of the investigation and be subjected to further scrutiny by law enforcement with varying degrees, based on the search approach. During genealogy investigations, law enforcement may also contact people who are not in the database and clearly not the potential known donor of the sample, to

acquire samples for DNA testing in order to narrow investigative leads and reduce investigator workload [31]. While contacted individuals can decide whether to cooperate when contacted by law enforcement to consensually provide a DNA sample, some individuals may feel coerced or intimidated simply by the requests. Moreover, concerns exist about the possibility of surreptitious collections of samples.

Profiles and other data in forensic STR databases are managed and maintained by government agencies under strict privacy protections [89] and policies regarding the sharing of that data with authorized agencies. Some propose that these searching approaches each bring varying degrees of additional privacy and intrusion concerns [29,75,77,90]. It has been suggested that government databases were not intended for these kinship type familial and Y-STR database searches [91] as is done with familial searching and the Y-STR database search concepts. Others assert that the language in the DNA Identification Act [83] is "for law enforcement identification purposes" and does not specify applicability to direct searches only. Genetic profiles in the genealogy databases are owned and managed by private companies or nonprofit organizations, and these data may be available online or stored on the cloud. Users may have the option to opt in or out for law enforcement searches, depending on the policies of the databases. A recent review of the policies of 22 genealogy companies and databases showed that only 4 provided their policies on law enforcement agencies access to their services [71]. The limits and consequences of privacy protection are, thus, very different between forensic databases and IGG databases.

Another propounded issue is the notion that certain populations are overrepresented in the government databases, which results in genetic discrimination [92,93]. Anecdotal evidence thus far does not seem to support this position, and more studies are needed. From a population component perspective, IGG targets individuals of European descent [27,75]. However, familial searching and the Y-STR approach are limited to searching offender databases, whereas IGG currently accesses nongovernmental databases. Theoretically, with all these approaches additional information must be sought to confirm or refute a potential donor of a crime scene sample, and A-STR typing of the alleged donor and the crime scene sample should be performed, compared and concordant, where applicable.

It is also important to point out that forensic A-STR and Y-STR data provide little or no association with any known medical conditions or disease status [94]. These markers together may infer some general ancestry information [53,95], which to a certain degree is associated with some physical traits. By contrast, high-density SNP profiles contain far more information that could infer physical traits, medical conditions, disease status, unknown family relationships, etc. GEDMatch even provides a function to infer if an individual's parents are related. To date, full profiles in the IGG databases are usually not disclosed to law enforcement, but only the relationships between a pair of profiles, possibly together with the length and location of the shared SNP blocks, may be provided to law enforcement. Still, a large proportion of the profiles in GEDMatch may be extracted by specially designed queries [84,85]. Security, privacy, access, and informed consent are certainly issues to be addressed with high-density SNP data.

Beyond genealogy and forensic databases, medical research projects have generated hundreds of thousands of WGS profiles. For example, Iceland has sequenced about half of its citizens for medical research [96]. The same data can technically be used to identify everyone in Iceland. These same databases are potential resources to help identify the source of a crime scene profile. Policies are needed to decide on whether these medical research data can be used for law enforcement purposes, and what corollary and necessary regulations are needed to protect these data.

Law enforcement agencies use other publicly accessible resources (e.g., social media accounts) in criminal investigations [97] and would seem to have the same access right to access public genealogy databases [98]. In this context, law enforcement analogizes the use of publicly available genetic data to identify the source of crime scene evidence to the intrusion level of other forms of evidence and level of intrusion, such as partial license plate information [99]. However, genetic data may be viewed differently as it is more personal and more intrusive. While there may not yet be case law addressing the use of these searching approaches, some states have proposed legislation to restrict that use, although no legislation has yet been passed [100,101].

The issues on privacy raised above are not intended to be exhaustive, and there are other topics which might be considered, such as any necessary differences in securing data based on source (victim and elimination data compared with the crime scene data versus suspect or of convicted felon information) and whether individuals have the right to determine the reach of his or her relatives through genetic testing. Regardless, an intrusion or invasion of privacy may be a proportionality issue. Society may be more willing to permit these searching methods for certain types of crimes, such as violent crimes and those committed against children, or for identifying missing persons [98]. Thus, a society's comfort level with intrusions will likely vary and will necessarily have to be addressed in accordance with the respective legal systems.

12 | LEGISLATION

Government forensic DNA databases legislative and policy restrictions are set by the respective countries. The US Congress passed the DNA Identification Act in 1994, which authorized the FBI to create a national DNA database and specified the requirements for participation in NDIS and the DNA data maintained in NDIS. The UK passed the Crime and Security Act in 2010, which allowed the establishment of a DNA database. The Criminal Procedure Law of China authorizes the generation of a DNA databases. Singapore also enacted its Registration of Criminals Act, which permitted databases. Over 60 countries have established DNA databases. These laws tend not to specify which specific DNA markers should be included in the databases. This may lend support for the conclusion that Y-STR markers can be allowed to

TABLE 1 Comparison of familial searching, Y-STR database searching, and IGG

	Familial Searching	Y-STR database searching[a]	IGG
Who provides the test?	Accredited forensic DNA laboratories	Accredited forensic DNA laboratories	DNA testing service providers: Ortham, DNA solutions, etc; Genealogy service providers: Bode, Parabon NanoLabs, DNA Doe Project, etc.
Genotyping technology	A-STR and Y-STR validated by forensic laboratories	A-STR and Y-STR validated by forensic laboratories	High-density SNP panels (microarray or whole-genome sequencing) not validated by forensic laboratories
Expertise	Accredited laboratories, DNA lab analysts	Accredited laboratories, DNA lab analysts	Service lab, genealogist, and DNA analysts
DNA quantity	500–1000 pg recommended; lower input may work	500–1000 pg recommended; lower input may work	At least 1 ng; Bode suggests >20 ng; Parabon NanoLabs suggests >10 ng
Database to search	Forensic A-STR database (e.g., NDIS)	Forensic Y-STR database	GEDmatch (~260 K opt-in profiles), Family tree DNA (~1 M profiles), DNASolves, etc.
Data quality	Verified profiles	Verified profiles	Unverified profiles
Data security	Secured	Secured	Less secure, more vulnerable to attacks; possible data leakage
Methods to measure relationship	Likelihood ratio; number of shared alleles	Number of mismatched steps	Total length of IBD segments; IBS, and combinations thereof
Relationships	1st degree relationships (parent–child, full sibling)	Any males in the same lineage	Can detect 3rd degree or more distant relationships, depending on the number of SNPs
# of successes as of now	100 s, mostly in United States and Europe	10 000 s, mostly in China	100 s, mostly in United States
Instrument cost	Existing instruments in DNA laboratories	Existing instruments in DNA laboratories	$100 000 to $1 000 000
Cost of DNA tests	$1000 s per case depending on how many Y-STR profiles to type during filtering	Build database: $30-50 per profile excluding labor cost, depending on workflow and the reagents; Per case: $100 s depending on how many A-STR profiles to type in the follow-up investigation	$3,000~$10,000 per case for IGG service provides; plus A-STR typing in the forensic laboratories
Time to results	Days to weeks	Days to weeks	Weeks to months mostly due to extensive non-DNA genealogy searching
Advantages	Existing database and technologies; validated and proficiency tested; existing legislation; safe and secured database	Existing technologies; existing legislation; validated and proficiency tested; safe and secured database; high efficiency search for male lineage; associate to surnames	Able to find distant relatives; able to find female relatives
Limitations	Large candidate list after searching; relies on Y-STR filtering	Only detect male lineage relatives; high upfront cost to build a Y-STR database if retyping all reference profiles; may need genealogists' assistance for distant matches	Not forensically validated technology; heavily relies on genealogists and public SNP databases; costly for routine analyses; data may not be secure; far more invasive private information generated than other methods
Legislation/ regulation	Allowed: ≥12 states in the United States, UK, Zealand, France, Netherland, Spain, China, etc. Prohibited: Maryland and Washington DC; Canada	Covered by current legislation for A-STR database	None

[a]Y-STR searching in the genetic genealogy database is more similar to IGG and thus is not included in the Y-STR database searching.

be entered into national DNA database systems, and thus far it appears that none of the countries building Y-STR databases have enacted specific laws permitting the establishment of their Y-STR databases.

Familial searching has been conducted in many countries, such as the UK, The Netherlands, Australia, New Zealand, and the United States, with or without policy or legislation, but Canada appears to require legislation [14,17]. Familial searching is performed in at least 12 US states but is prohibited in Maryland and Washington DC [14,17]. Interestingly, in the 12 states that make use of familial searching, none have legislation specifically stating the use of this approach.

Currently, there is no specific federal statutory legislation designed to regulate IGG and prohibit some practices. The profiles in GEDMatch were voluntarily uploaded by the individuals who own their profiles, and these genetic genealogy databases, unlike government databases, are unregulated. Individuals in FamilyTreeDNA's database needed to opt-out to restrict searching by law enforcement. It appears that the other DTC companies do not permit law enforcement to search the profiles contained in those databases. Recently, the US Department of Justice presented an interim policy on IGG, which was "designed to balance the Department's relentless commitment to solving violent crimes and protecting public safety against equally important public interests – such as preserving the privacy and civil liberties of all citizens" [47–49]. Yet, this policy, at least in part, may not be applicable to state and local jurisdictions. Legislation also has been introduced or proposed in multiple states across the United States, either banning (e.g., Utah, California) [100,101] or regulating IGG (e.g., Maryland, Washington, New York) (102).

13 | SUMMARY

To extend potential investigative leads in criminal cases, particularly violent crimes, and identify missing persons, searching DNA databases for possible relatives of the true source of a sample is an effective and successful approach. Three indirect investigation approaches for searching for relatives in DNA databases are addressed by this paper: familial searching, Y-STR database searching, and IGG. Table 1, herein, summarizes the details of these approaches from different perspectives. All three approaches are viable and should be considered as methods to increase the value of DNA databasing. Y-STR database searching and IGG are advantageous since they should be able to assist in more cases than familial searching readily identifying distant relatives. In contrast, familial searching can be performed more readily with existing laboratory systems and utilizing at least state-level databases. Every country or state may have its own unique economic, technical, cultural, and legal considerations and should decide the best approach(es) to fit those circumstances. Regardless of the approach, the ultimate goal should be the same: generate investigative leads and solve active and cold criminal cases to public safety, under stringent policies and security practices designed to protect the privacy of its citizenry.

ACKNOWLEDGMENT

The authors would like to acknowledge Dawn Moore Boswell, Christina Capt, Rock Harmon, Brian Hoey, Colleen Fitzpatrick, Christopher Syn, SallyAnn Harbison, and Carole Field for their assistance in providing information on their work.

ORCID

Jianye Ge https://orcid.org/0000-0001-8724-075X

REFERENCES

1. Federal Bureau of Investigation. CODIS - NDIS statistics. https://www.fbi.gov/services/laboratory/biometric-analysis/codis/ndis-statistics. Accessed 11 Sept 2020.
2. GEDMatch. https://www.gedmatch.com/. Accessed 11 Sept 2020.
3. Miller G. Familial DNA testing scores a win in serial killer case. Science. 2010;329(5989):262. https://doi.org/10.1126/science.329.5989.262
4. Xinhua News Agency. Distant cousin's DNA locked up the suspect, Gao Chengyong. http://www.xinhuanet.com/politics/2016-08/30/c_129261666.htm. Accessed 11 Sept 2020.
5. New York Times. Man thought to be China's Jack the Ripper Is arrested. https://www.nytimes.com/2016/08/30/world/asia/china-jack-the-ripper-gao-chengyong.html. Accessed 11 Sept 2020.
6. Kaiser J. We will find you: DNA search used to nab Golden State Killer can home in on about 60% of white Americans. Science. 2018. https://doi.org/10.1126/science.aav7021.
7. Kayser M. Forensic use of Y-chromosome DNA: a general overview. Hum Genet. 2017;136(5):621–35. https://doi.org/10.1007/s00439-017-1776-9.
8. Evett IW. Evaluating DNA profiles in a case where the defence is "It was my brother". J Forensic Sci Soc. 1992;32(1):5–14. https://doi.org/10.1016/s0015-7368(92)73041-2.
9. Bieber FR, Brenner CH, Lazer D. Human genetics. Finding criminals through DNA of their relatives. Science. 2006;312(5778):1315–6. https://doi.org/10.1126/science.1122655.
10. Bureau of Justice Statistics. Correctional populations in United States, 1996. Washington, DC: U.S. Department of Justice, Office of Justice Programs; 1996. p. 62.
11. Ge J, Budowle B. Kinship index variations among populations and thresholds for familial searching. PLoS One. 2012;7(5):e37474. https://doi.org/10.1371/journal.pone.0037474.
12. Myers SP, Timken MD, Piucci ML, Sims GA, Greenwald MA, Weigand JJ, et al. Searching for first-degree familial relationships in California's offender DNA database: validation of a likelihood ratio-based approach. Forensic Sci Int Genet. 2011;5(5):493–500. https://doi.org/10.1016/j.fsigen.2010.10.010.
13. Ge J, Chakraborty R, Eisenberg A, Budowle B. Comparisons of familial DNA database searching strategies. J Forensic Sci. 2011;56(6):1448–56. https://doi.org/10.1111/j.1556-4029.2011.01867.x.
14. Anderson J, Strang L, Matthies C, Piquado T. Forensic familial and moderate stringency DNA searches: policies and practices in the United States, England, and Wales. Santa Monica, CA: RAND Corporation; 2019.
15. NBC News. New DNA technology helps solve Santa Ana child rape case, officials say. https://www.nbclosangeles.com/news/local/dna-technology-solves-santa-ana-child-rape-case/1891699/. Accessed 11 Sept 2020.

16. Granja R, Machado H. Ethical controversies of familial searching: The views of stakeholders in the United Kingdom and in Poland. Sci Technol Hum Values. 2019;44(6):1068–92. https://doi.org/10.1177/0162243919828219.
17. García Ó, Crespillo M, Yurrebaso I. Suspects identification through "familial searching" in DNA databases of criminal interest. Social, ethical and scientific implications. Span J Leg Med. 2017;43(1):26–34. https://doi.org/10.1016/j.remle.2017.02.002.
18. RTI International. Familial DNA searching: current approaches. Research Triangle Park, NC: RTI International; 2015.
19. Sanlian Life magazine. How was the murderer of the Southern Medical University 28 years ago found in three days? https://finance.sina.cn/2020-02-26/detail-iimxxstf4616946.d.html. Accessed 11 Sept 2020.
20. Liu H, Li X, Mulero J, Carbonaro A, Short M, Ge J. A convenient guideline to determine if two Y-STR profiles are from the same lineage. Electrophoresis. 2016;37(12):1659–68. https://doi.org/10.1002/elps.201500566.
21. Schellberg T. Announcement of the 4th DNA hit of the year. Human Identification Solutions (HIDS) Virtual Conference. 2020. https://wwwthermofishercom/us/en/home/industrial/forensics/human-identification/human-identification-solutions-conference-2020.html. Accessed 5 Oct 2020.
22. Ge J, Sun H, Li H, Liu C, Yan J, Budowle B. Future directions of forensic DNA databases. Croat Med J. 2014;55(2):163–6. https://doi.org/10.3325/cmj.2014.55.163.
23. Willuweit S, Roewer L. The new Y chromosome haplotype reference database. Forensic Sci Int Genet. 2015;15:43–8. https://doi.org/10.1016/j.fsigen.2014.11.024.
24. Promega. Under the microscope – Colleen Fitzpatrick. https://www.ishinews.com/under-the-microscope-colleen-fitzpatrick/. Accessed 11 Sept 2020.
25. Investigation Discovery. DNA test stunningly accurate as serial hammer slayer with predicted last name arrested. https://www.investigationdiscovery.com/crimefeed/id-shows/dna-test-stunningly-accurate-as-serial-hammer-slayer-with-predicted-last-name-arrested. Accessed 11 Sept 2020.
26. KOTA TV.Rapid City police solve a cold case after 51 years. https://www.kotatv.com/content/news/Rapid-City-Police-solve-a-cold-case-51-years-later-511436101.html. Accessed 11 Sept 2020.
27. Erlich Y, Shor T, Pe'er I, Carmi S. Identity inference of genomic data using long-range familial searches. Science. 2018;362(6415):690–4. https://doi.org/10.1126/science.aau4832.
28. Ge J, Budowle B. How many familial relationship testing results could be wrong? PLoS Genet. 2020;16(8):e1008929. https://doi.org/10.1371/journal.pgen.1008929.
29. Wickenheiser RA. Forensic genealogy, bioethics and the Golden State Killer case. Forensic Sci Int. 2019;1:114–25. https://doi.org/10.1016/j.fsisyn.2019.07.003.
30. Katsanis SH. Pedigrees and perpetrators: Uses of DNA and genealogy in forensic investigations. Annu Rev Genomics Hum Genet. 2020;21:535–64. https://doi.org/10.1146/annurev-genom-111819-084213.
31. Greytak EM, Moore C, Armentrout SL. Genetic genealogy for cold case and active investigations. Forensic Sci Int. 2019;299:103–13. https://doi.org/10.1016/j.forsciint.2019.03.039.
32. Bettinger BT, Perl J.The shared cM oroject 3.0 tool v4. 2018. https://dnapainter.com/tools/sharedcmv4. Accessed 11 Sept 2020.
33. Verogen. GEDmatch partners with genomics firm. https://Verogen.com/gedmatch-partners-with-genomics-firm/. Accessed 11 Sept 2020.
34. Washington Business Journal. The Reston company cracking cold police cases is coming to your TV. https://www.bizjournals.com/washington/news/2020/05/06/the-reston-company-cracking-cold-police-cases-is.html. Accessed 11 Sept 2020.
35. Thomson J, Clayton T, Cleary J, Gleeson M, Kennett D, Leonard M, et al. The effectiveness of forensic genealogy techniques in The united kingdom – an experimental assessment. Forensic Sci Int Genet Suppl Ser. 2019;7(1):765–7. https://doi.org/10.1016/j.fsigss.2019.10.169.
36. Tillmar A, Sjolund P, Lundqvist B, Klippmark T, Algenas C, Green H. Whole-genome sequencing of human remains to enable genealogy DNA database searches - A case report. Forensic Sci Int Genet. 2020;46:102233. https://doi.org/10.1016/j.fsigen.2020.102233.
37. Samuel G, Kennett D. The impact of investigative genetic genealogy: perceptions of UK professional and public stakeholders. Forensic Sci Int Genet. 2020;48:102366. https://doi.org/10.1016/j.fsigen.2020.102366.
38. Liu J, Ji A, Li C, Wang G. Research progress of forensic genealogy analysis. Forensic Science and Technology. 2019;44(3):189–94. https://doi.org/10.16467/j.1008-3650.2019.03.001.
39. Zhu B, Guo Y, Li S, Liu C, Cong B. New genetic sleuthing tool: Forensic genealogy. Chin Sci Bull. 2019;64(22):2274–2278. https://doi.org/10.1360/N972019-00266.
40. Wang DY, Gopinath S, Lagacé RE, Norona W, Hennessy LK, Short ML, et al. Developmental validation of the GlobalFiler® Express PCR Amplification Kit: A 6-dye multiplex assay for the direct amplification of reference samples. Forensic Sci Int Genet. 2015;19:148–55. https://doi.org/10.1016/j.fsigen.2015.07.013.
41. Gopinath S, Zhong C, Nguyen V, Ge J, Lagacé RE, Short ML, et al. Developmental validation of the Yfiler® Plus PCR Amplification Kit: An enhanced Y-STR multiplex for casework and database applications. Forensic Sci Int Genet. 2016;24:164–75. https://doi.org/10.1016/j.fsigen.2016.07.006.
42. Ensenberger MG, Lenz KA, Matthies LK, Hadinoto GM, Schienman JE, Przech AJ, et al. Developmental validation of the PowerPlex® fusion 6C system. Forensic Sci Int Genet. 2016;21:134–44. https://doi.org/10.1016/j.fsigen.2015.12.011.
43. Jäger AC, Alvarez ML, Davis CP, Guzmán E, Han Y, Way L, et al. Developmental validation of the MiSeq FGx Forensic Genomics System for targeted next generation sequencing in forensic DNA casework and database laboratories. Forensic Sci Int Genet. 2017;28:52–70. https://doi.org/10.1016/j.fsigen.2017.01.011.
44. Alonso A, Barrio PA, Muller P, Kocher S, Berger B, Martin P, et al. Current state-of-art of STR sequencing in forensic genetics. Electrophoresis. 2018;39(21):2655–68. https://doi.org/10.1002/elps.201800030.
45. Dagnall CL, Morton LM, Hicks BD, Li S, Zhou W, Karlins E, et al. Successful use of whole genome amplified DNA from multiple source types for high-density Illumina SNP microarrays. BMC Genom. 2018;19(1):182. https://doi.org/10.1186/s12864-018-4572-6.
46. Bode Technology. Bode FGS - DNA testing & analysis frequently asked questions (FAQs). https://www.bodetech.com/frequently-asked-questions/109. Accessed 23 Jun 2020.
47. Department of Justice. Department of Justice announces interim policy on emerging method to generate leads for unsolved violent crimes. https://www.justice.gov/opa/pr/department-justice-announces-interim-policy-emerging-method-generate-leads-unsolved-violent. Accessed 11 Sept 2020.
48. United States Department of Justice. Interim policy, forensic genetic genealogical DNA analysis and searching. 2019. https://www.justice.gov/olp/page/file/1204386/download. Accessed 23 Jun 2020.
49. Callaghan TF. Responsible genetic genealogy. Science. 2019;366(6462):155. https://doi.org/10.1126/science.aaz6578.
50. Karantzali E, Rosmaraki P, Kotsakis A, Le Roux-Le Pajolec MG, Fitsialos G. The effect of FBI CODIS Core STR Loci

expansion on familial DNA database searching. Forensic Sci Int Genet. 2019;43:102129. https://doi.org/10.1016/j.fsigen.2019.07.008.
51. Thermo Fisher Scientific. How to determine if two Y-STR haplotypes are from the same male lineage. https://www.thermofisher.com/blog/behindthebench/how-to-determine-if-two-y-str-haplotypes-are-from-the-same-male-lineage/. Accessed 11 Sept 2020.
52. Ge J, Budowle B, Aranda XG, Planz JV, Eisenberg AJ, Chakraborty R. Mutation rates at Y chromosome short tandem repeats in Texas populations. Forensic Sci Int Genet. 2009;3(3):179–84. https://doi.org/10.1016/j.fsigen.2009.01.007.
53. Ge J, Budowle B, Planz JV, Eisenberg AJ, Ballantyne J, Chakraborty R. US forensic Y-chromosome short tandem repeats database. Leg Med. 2010;12(6):289–95. https://doi.org/10.1016/j.legalmed.2010.07.006.
54. Ballantyne KN, Keerl V, Wollstein A, Choi Y, Zuniga SB, Ralf A, et al. A new future of forensic Y-chromosome analysis: rapidly mutating Y-STRs for differentiating male relatives and paternal lineages. Forensic Sci Int Genet. 2012;6(2):208–18. https://doi.org/10.1016/j.fsigen.2011.04.017.
55. Ge J, Eisenberg A, Yan J, Chakraborty R, Budowle B. Pedigree likelihood ratio for lineage markers. Int J Legal Med. 2011;125(4):519–25. https://doi.org/10.1007/s00414-010-0514-9.
56. Ball CA, Barber MJ, Byrnes J, Carbonetto P, Chahine KG, Curtis RE, et al. Ancestry DNA matching white paper. Technical report. AncestryDNA. 2016. https://www.ancestry.com/corporate/sites/default/files/AncestryDNA-Matching-White-Paper.pdf. Accessed 5 Oct 2020.
57. Henn BM, Hon L, Macpherson JM, Eriksson N, Saxonov S, Pe'er I, et al. Cryptic distant relatives are common in both isolated and cosmopolitan genetic samples. PLoS One. 2012;7(4):e34267. https://doi.org/10.1371/journal.pone.0034267.
58. Manichaikul A, Mychaleckyj JC, Rich SS, Daly K, Sale M, Chen WM. Robust relationship inference in genome-wide association studies. Bioinformatics. 2010;26(22):2867–73. https://doi.org/10.1093/bioinformatics/btq559.
59. Al-Khudhair A, Qiu S, Wyse M, Chowdhury S, Cheng X, Bekbolsynov D, et al. Inference of distant genetic relations in humans using "1000 genomes". Genome Biol Evol. 2015;7(2):481–92. https://doi.org/10.1093/gbe/evv003.
60. DNA Resource. Global DNA databases. http://www.dnaresource.com/resources.html. Accessed 10 Sept 2020.
61. National DNA database statistics. https://www.gov.uk/government/statistics/national-dna-database-statistics. Accessed 23 Jun 2020.
62. Hares DR. Selection and implementation of expanded CODIS core loci in the United States. Forensic Sci Int Genet. 2015;17:33–4. https://doi.org/10.1016/j.fsigen.2015.03.006.
63. Phillips C. A genomic audit of newly-adopted autosomal STRs for forensic identification. Forensic Sci Int Genet. 2017;29:193–204. https://doi.org/10.1016/j.fsigen.2017.04.011.
64. National Institute of Standards and Technology. Core STR loci used in human identity testing. https://strbase.nist.gov/coreSTRs.htm. Accessed 11 Sept 2020.
65. Budowle B, Moretti TR, Niezgoda SJ, Brown BL, editors. CODIS and PCR-based short tandem repeat loci: law enforcement tools. Proceedings of the Second European Symposium on Human Identification; 1998 June 10–12; Innsbruck, Austria. Madison, WI: Promega Corporation, 1998.
66. Budowle B, Ge J, Aranda XG, Planz JV, Eisenberg AJ, Chakraborty R. Texas population substructure and its impact on estimating the rarity of Y STR haplotypes from DNA evidence. J Forensic Sci. 2009;54(5):1016–21. https://doi.org/10.1111/j.1556-4029.2009.01105.x.
67. Ge JY, Yan JW, Xie Q, Sun HY, Zhou HG, Li B. Development of Chinese forensic Y-STR DNA database. Fa Yi Xue Za Zhi. 2013;29(3):212–5, 21.
68. Snyder HN. Arrest in the United States, 1990–2010. Washington, DC: Bureau of Justice Statistics, Office of Justice Programs, US Department of Justice; 2012.
69. Alexia Cooper ELS. Homicide Trends in the United States, 1980–2008. Washington, DC: Bureau of Justice Statistics, Office of Justice Programs, US Department of Justice; 2011.
70. Regalado A. More than 26 million people have taken an at-home ancestry test. MIT Technology Review. 2019. https://www.technologyreview.com/2019/02/11/103446/more-than-26-million-people-have-taken-an-at-home-ancestry-test/. Accessed 5 Oct 2020.
71. Skeva S, Larmuseau MH, Shabani M. Review of policies of companies and databases regarding access to customers' genealogy data for law enforcement purposes. Per Med. 2020;17(2):141–53. https://doi.org/10.2217/pme-2019-0100.
72. BuzzFeed News. One of the biggest at-home DNA testing companies is working with the FBI. https://www.buzzfeednews.com/article/salvadorhernandez/family-tree-dna-fbi-investigative-genealogy-privacy. Accessed 10 Sept 2020.
73. Ram N, Roberts JL. Forensic genealogy and the power of defaults. Nat Biotechnol. 2019;37(7):707–8. https://doi.org/10.1038/s41587-019-0172-5.
74. FamilyTreeDNA privacy statement. https://www.familytreedna.com/legal/privacy-statement. Accessed 23 June 2020.
75. Ram N, Guerrini CJ, McGuire AL. Genealogy databases and the future of criminal investigation. Science. 2018;360(6393):1078–9. https://doi.org/10.1126/science.aau1083.
76. Greytak EM, Moore C, Armentrout SL. RE: Identity inference of genomic data using long-range familial searches. Erlich et al. Science 2018;362(6415):690–4. Science. eLetter. 2018. https://science.sciencemag.org/content/362/6415/690/tab-e-letters. Accessed 5 Oct 2020.
77. Syndercombe Court D. Forensic genealogy: Some serious concerns. Forensic Sci Int Genet. 2018;36:203–4. https://doi.org/10.1016/j.fsigen.2018.07.011.
78. Ellenbogen P, Narayanan A. Identification of anonymous DNA using genealogical triangulation. bioRxiv. 2019. https://doi.org/10.1101/531269.
79. Gymrek M, McGuire AL, Golan D, Halperin E, Erlich Y. Identifying personal genomes by surname inference. Science. 2013;339(6117):321–4. https://doi.org/10.1126/science.1229566.
80. Budowle B, Masibay A, Anderson SJ, Barna C, Biega L, Brenneke S, et al. STR primer concordance study. Forensic Sci Int. 2001;124(1):47–54. https://doi.org/10.1016/S0379-0738(01)00563-1.
81. Davis C, Ge J, King J, Malik N, Weirich V, Eisenberg AJ, et al. Variants observed for STR locus SE33: a concordance study. Forensic Sci Int Genet. 2012;6(4):494–7. https://doi.org/10.1016/j.fsigen.2011.12.002.
82. Hong H, Xu L, Liu J, Jones WD, Su Z, Ning B, et al. Technical reproducibility of genotyping SNP arrays used in genome-wide association studies. PLoS One. 2012;7(9):e44483. https://doi.org/10.1371/journal.pone.0044483.
83. DNA Identification Act. https://www.congress.gov/bill/103rd-congress/house-bill/829/text. Accessed 11 Sept 2020.
84. Edge MD, Coop G. Attacks on genetic privacy via uploads to genealogical databases. Elife. 2020;9:e51810. https://doi.org/10.7554/eLife.51810.
85. Ney P, Ceze L, Kohno T. Genotype extraction and false relative attacks: security risks to third-party genetic genealogy services beyond identity inference. Preprint Posted. 2020;10(29):19.

86. MarketWatch. MyHeritage hack affects 92 million customers, reveals more risks with genealogy sites. https://www.MarketWatch.com/story/myheritage-dna-site-hack-reveals-risks-associated-with-genealogy-sites-2018-06-06. Accessed 10 Sept 2020.
87. TechCrunch. GEDmatch confirms data breach after users' DNA profile data made available to police. https://TechCrunch.com/2020/07/22/gedmatch-investigating-dna-profile-law-enforcement/. Accessed 11 Sept 2020.
88. CBS News. Private equity wants to own your DNA. https://www.cbsnews.com/news/blackstone-private-equity-ancestry-com-dna/. Accessed 11 Sept 2020.
89. Federal Bureau of Investigation. Freedom of Information/Privacy Act. https://www.fbi.gov/services/information-management/foipa Accessed 11 Sept 2020.
90. Greytak EM, Kaye DH, Budowle B, Moore C, Armentrout SL. Privacy and genetic genealogy data. Science. 2018;361(6405):857. https://doi.org/10.1126/science.aav0330.
91. Federal Bureau of Investigation. Frequently asked questions on CODIS and NDIS. https://www.fbi.gov/services/laboratory/biometric-analysis/codis/codis-and-ndis-fact-sheet. Accessed 11 Sept 2020.
92. Murphy E, Tong J. The racial composition of forensic DNA databases. Calif Law Rev. 2019;108:19–54. https://doi.org/10.2139/ssrn.3477974.
93. Sokhansanj BA. Beyond protecting genetic privacy: understanding genetic discrimination through its disparate impact on racial minorities. Columbia Journal of Race and Law. 2012;2:279.
94. Hares DR. Expanding the CODIS core loci in the United States. Forensic Sci Int Genet. 2012;6(1):e52–e4. https://doi.org/10.1016/j.fsigen.2011.04.012.
95. Graydon M, Cholette F, Ng L-K. Inferring ethnicity using 15 autosomal STR loci—Comparisons among populations of similar and distinctly different physical traits. Forensic Sci Int Genet. 2009;3(4):251–4. https://doi.org/10.1016/j.fsigen.2009.03.002.
96. McClatchy. This nation faces a DNA dilemma: Whether to notify people carrying cancer genes. https://www.McClatchy.c.com/news/nation-world/article213014904.html. Accessed 11 Sept 2020.
97. Kim K, Mohr AO-NE. 2016 Law enforcement use of social media survey. A Joint Publication by the International Association of Chiefs of Police and the Urban Institute. Washington, DC: Urban Institute; 2017. p. 1–22.
98. Guerrini CJ, Robinson JO, Petersen D, McGuire AL. Should police have access to genetic genealogy databases? Capturing the Golden State Killer and other criminals using a controversial new forensic technique. PLoS Biol. 2018;16(10):e2006906. https://doi.org/10.1371/journal.pbio.2006906.
99. Gierlack K, Williams S, LaTourrette T, Anderson JM, Mayer LA. License plate readers for law enforcement: Opportunities and obstacles. Santa Monica, CA: Rand Corporation; 2014.
100. State of Utah. H.B. 231 Genetic Information Amendments. https://le.utah.gov/~2020/bills/static/HB0231.html. Accessed 23 Jun 2020.
101. The Sacramento Bee. Who's looking at your DNA data? California lawmaker introduces genetic privacy bill. https://www.sacbee.com/news/politics-government/capitol-alert/article240227106.html. Accessed 11 Sept 2020.
102. The PEW Charitable Trusts. DNA databases are boon to police but menace to privacy, critics say. https://www.pewtrusts.org/en/research-and-analysis/blogs/stateline/2020/02/20/dna-databases-are-boon-to-police-but-menace-to-privacy-critics-say. Accessed 11 Sept 2020.

How to cite this article: Ge J, Budowle B. Forensic investigation approaches of searching relatives in DNA databases. *J Forensic Sci*. 2020;00:1–14. https://doi.org/10.1111/1556-4029.14615